

Deterministic by design. Auditable by default.

NextFlow records every phase of an automation pipeline — snapshot, scan, simulate, relay — as an immutable trace.

Replay-safe. Dependency-checked. Fully accountable, end to end.

6 PHASES · 1 ISOLATION GATE · 0 SILENT FAILURES

● LIVE TRACE

INSTANCE 986b3036 · P

- P0 ☒ Snapshot
- P1 ☒ Scan + Analysis
- GATE ☒ Isolation check
- P2 ☐ Pre-simulation
- P3 ☐ Simulation
- P4 ☐ Inspection

The Trust Problem in Modern Pipelines

Manual detection. Blind fixes. Lost documentation. The hidden cost of non-deterministic automation.



Silent Failures

Errors that propagate downstream without detection until it's too late.



Blind Execution

Applying changes directly to production without a verified simulation state.

● FAILED TRACE

INSTANCE fa82c192

P0  Snapshot

P1  **Scan + Analysis [CRITICAL]**

GATE  Isolation
check

P2  Pre-
simulation

THE NEW STANDARD

The **Deterministic** Solution

Work on a candidate copy. Verify with mathematical certainty. Then relay to production.

STEP 01

Capture State

Create an immutable snapshot of the current production environment.



STEP 02

Simulate Fixes

Apply mutations on an isolated candidate copy without affecting live users.



STEP 03

Verified Relay

User-reviewed transition from candidate to production after full audit.

The Six Deterministic Phases

PHASE 00



Snapshot

Capture immutable state of the target environment before any action.

PHASE 01



Scan & Analysis

Static analysis and integrity check of all pipeline components.

PHASE 02

Pre-simulation

Dependency resolution and scoring of simulation readiness.

PHASE 03



Simulation

Deterministic execution on isolated candidate copies.

PHASE 04



Inspection

Automated hash verification and integrity audit of simulation output.

PHASE 05

Relay

Final transparency gate for user-reviewed transition to production.

Phase 0 & 1:

Snapshot & Scan

Establishing the immutable ground truth before any mutation occurs.



Immutable Capture

Phase 0 creates a point-in-time snapshot of the entire environment state.



Integrity Analysis

Phase 1 scans all components for structural integrity and dependency drift.

```
nextflow-scan --target ./prod
```

```
[ecf24601] core/engine.py VERIFIED
```

```
[2c9e649e] pipeline/state.py VERIFIED
```

```
[a8f1b2c3] api/routes.py VERIFIED
```

```
[7d8e9f0a] utils/auth.py VERIFIED
```

```
> Snapshot P0 complete. 142 files captured.
```

```
> Scan P1 complete. 0 integrity issues found.
```

— THE GATEKEEPER

Phase 2 & Gate: The Threshold

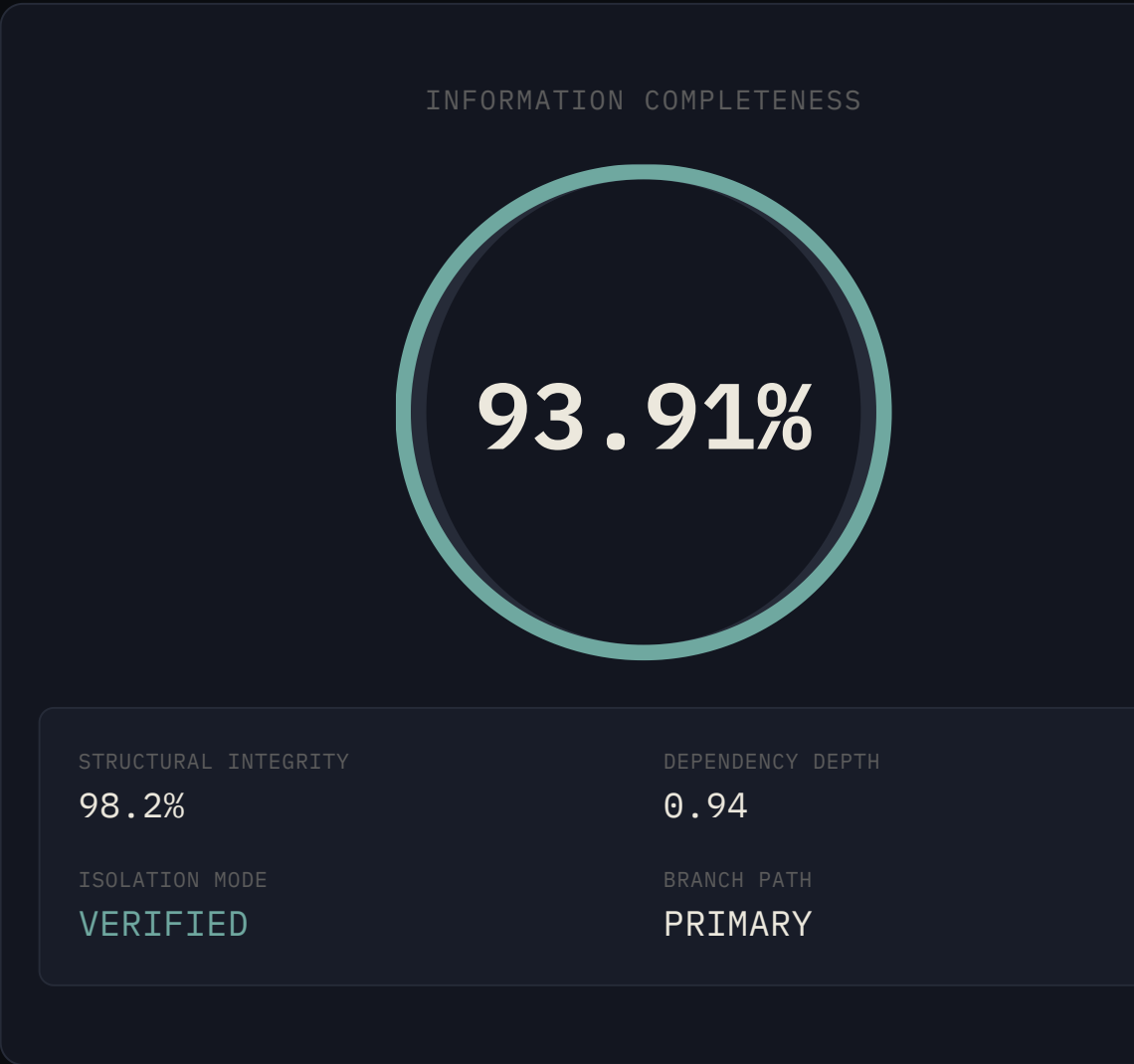
Quantifying certainty before execution. No action is taken without verified completeness.

\$ Deterministic Scoring

Phase 2 calculates an integrity score based on structural analysis.

🔒 The 93.91% Gate

A mathematically defined threshold that determines the isolation strategy.



Phase 3 & 4: Simulation & Inspection

Verifying the delta with mathematical precision. Every mutation is audited before it's earned.



Isolated Simulation

Phase 3 applies mutations on isolated candidate copies for zero-risk testing.



Integrity Inspection

Phase 4 audits the simulation output through automated hash verification.

DIFF: module_ambiguous_import.py

```
12 def process_edge_routing(data):  
13 -     from .utils import route_resolver  
13 +     from pipeline.core.isolation import route_resolver  
14 +     # Resolved ambiguous import via Phase 3 Simulation  
15     return route_resolver(data)
```

> Hash verified: 2c9e649ee75143ad [OK]

— VISUALIZING THE PIPELINE

Interactive Control Room Demo

Showcasing the live dashboard and real-time execution trace. See the deterministic engine in action before you commit.



Real-Time Visibility

Watch the pipeline progress through all 6 phases with instant feedback.



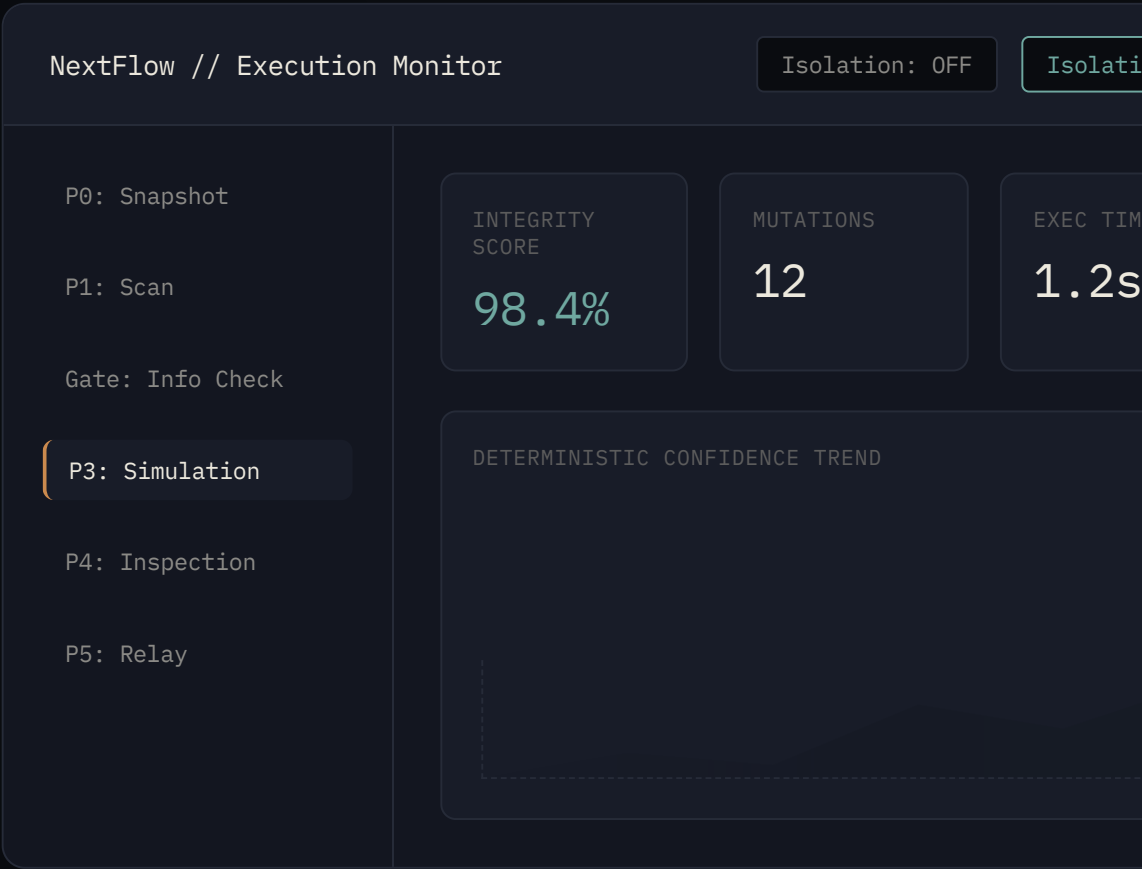
Interactive Isolation Toggle

Switch between isolated and live execution modes with a single click.



Phase-by-Phase Inspection

Drill down into specific AST diffs and deterministic scoring metrics.



NEXTFLOW

THE FUTURE OF PIPELINE INTEGRITY

Every Output is Earned.

Join the era of deterministic software engineering. Zero
silent failures. Absolute transparency.

EST. 2026

nextflow.ai/launch

ACCESS THE CONTROL ROOM

VERSION 1.0.4